A fantastic title about minihalos that somehow grips a viewers attention for over 100 pages of detail

by

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Abstract

A truly brilliant abstract that captures the nuance of what I have done, without being overly detailed and dense so that a reader throws it out the window.

Acknowledgments

People who helped me during my Thesis: Yvette, CASA people, sibling.

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Chapter 1

Introduction

When looking up into the sky on a clear night those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these form galaxy clusters, with tens to thousands of individual galaxies becoming gravitationally bound to a single massive structure (somewhat analagous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the universe, with total masses of order $10^{14}-10^{15}\mathrm{M}_{\odot}$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that majority of the cluster mass to be from the innumerable galaxies held within, however, only a few percent of the total cluster mass comes this baryonic matter (stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7-10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and sparsest at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). However, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that was a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim 3\%$ of the mass budget taken by galaxies, $\sim 12\%$ from the ICM, and a staggering $\sim 85\%$ from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Long filaments of galaxies span across the space between clusters, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5-10^7 K. These

[1] Smaller clusters with only tens of galaxies are often known as groups.

[2] For those unaccustomed to astrophysical units, $\mathrm{M}_{\odot}$ is the solar mass and defined as $1\ \mathrm{M}_{\odot} \approx 10^{30}$ kg. A parsec is defined as $1\ \mathrm{pc} \approx 3 \times 10^{16}$ m. These unusual units are far more suited at the scale of the universe.

filaments are larger than clusters, though not gravitationally bound, and create a cosmic web in the sky with galaxy clusters located in the intersections of the web. This cosmic web is the largest structure in the universe, a vast network of dark matter that we can trace via the observable galaxies and clusters (Bond et al., 1996; Pimbblet, 2005) with ongoing work in detecting these filaments observationally (Tudorache et al., 2025). Clusters are understood to then form in part due to the accretion at these intersections of the WHIM and the cosmic web, alonside the more dominant mergers of smaller clusters. These highly energetic merger events create shocks and turbulence within the cluster environment, causing the ICM to heat.

Add references to merger stuff here.

An important distinction should be made when analysing clusters, between "disturbed" clusters undergoing merger events or still in a turbulent state from a recent merger, and "relaxed" clusters which have settled into an undisturbed state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures, and therefore, simpler modelling (Mantz et al., 2016). Many relaxed clusters have also been found to feature a "cool core", where the temperature at the centre of the cluster is significantly lower than the region outside the core. The high density of the ICM in the core causes a large outflux of X-ray emission, great enough to lower the cooling time below the expansion rate of the universe (known as Hubble time). In a closed system, this develops into a cooling flow causing the core temperature to drop and gas flows inwards (Peterson and Fabian, 2006; Hudson et al., 2010). Observational evidence supports this theory, however the magnitude of observed gas is much lower than theorised, implying the existence of a heating mechanism to counteract the cooling flow. One such source are radio galaxies which are active galactic nuclei (AGN) that emit radio synchrotron emission. This emission create jets and lobes that have been observed to interact with the ICM, distributing energy in "radio-mode" feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6 \times 10^{14}$ $M_\odot$)[3] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Less common in lower mass clusters, and unobserved within non-cool cores, minihalos are diffuse radio sources with currently unknown origins. This radio emission is localised within the cool cores, with radii from 50 kpc to $0.2R_{500}$.[4] This emission cannot be directly formed by radio sources such as AGN, as there is no observed spatial or spectral connection at any resolution or frequencies studied thus far. Furthermore, the spatial extent of these halos is well beyond the expected extent for particles sourced from an AGN (e.g. Bagchi et al., 2002; Giacintucci et al., 2014). Understanding the origins and physics behind minihalos is an ongoing process, with viable production methods proposed, and a lack of observational evidence to differentiate between them.

[3]The value M_{500} refers to the total mass within a radius of R_{500} where the clusters average enclosed density is 500 times the critical density of the universe at the cluster redshift. The critical density is a cosmological quantity, that has varyied over the lifetime of the universe.

[4]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius, the cooling, AGN feedback, and similar processes become significant to the behaviour of the ICM.

Working towards improving our understanding of minihalos will be the focus of the thesis. **Improve the ending of this section.**

1.1 Observations of Galaxy Clusters

The question that should then be asked is, how *do* we know all of this information about galaxy clusters? How are we able to measure the temperature, density, pressures within these clusters without travelling the impossibly vast distances to get there? The answer to this question is simple, by observing the radiation emitted from these structures. Different processes within the cluster, and more specifically ICM, provide information to the physical environment. This radiation is emitted across the electromagnetic spectrum and we design telescopes to capture and relay different bands in order to identify different components of what we observe. The question then becomes, how do observations at different frequency bands inform us to the structure and processes occuring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. Although incredibly informing about the stars and galaxies themselves, in the context of clusters, optical light has limited use. One usage is the ability to calculate the radial velcoity of cluster galaxies, which can then be used to inform stuff about mass. The other usage is to combine it with radio or X-ray data in order to determine what the source of emission is. For example does a radio burst have an optical counterpart? Then its likely a galaxy instead of some floating relativistic gas.

1.1.2 X-ray

The hot gas of the intracluster medium forms a thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, and photon energies between 0.1 keV and 100 keV.[5] The energy of the photons is dependent on the temperature of the ICM, resulting in most emissions being in the narrower range of 1 to 10 keV (Schneider, 2014).

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, however, this becomes limited at higher temperatures as the ionisation fraction increases. The distribution of line emissions can also be an indicator of abundance of elements such as C, N Ne, Mg, Si, S, Ar, Ca, and Fe. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total emissions can be dominated by line emission (Schneider, 2014).

[5] X-ray emission is historically measured in kilo-electron volts (energy) rather than frequency or wavelength.

Continuum radiation is due to brehmsstrahlung, or free-free radiation, which is where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection causes an acceleration induces a change in kinetic energy which is emitted as a photon. The spectral energy distribution of these emissions is related to the kinetic energy of the particles, and therefore, the temperature of the ICM. The scale of emissions is similarly impacted by the number of particles, and thereby the density of the ICM ρ_g in a given region. This gives rise to the following relationship to total continuum emissivity (Schneider, 2014).

$$\varepsilon_c \propto T^{\frac{1}{2}} \rho_g^2 \tag{1.1}$$

X-ray spectroscopy is a powerful tool to characterise the ICM and a clusters dynamical state. In general, telescopes trade off between high spatial or spectral resolutions, i.e. between detecting exactly where a photon originated from or an accurate measurement of the photon energy. This becomes somewhat of a constraint when observing galaxy clusters as we require reasonable spatial and spectral resolutions for these large objects. One common instrument is the space telescope *Chandra X-ray Observatory* launched in 1999 . Many studies of minihalos use *Chandra* observations to map the structure of the cool core and investigate potential relationships to the radio emission (e.g. Giacintucci et al., 2024). However, due to the aging detectors on *Chandra*, new observations aren't as good idk. A new telescope *XRISM* was launched in late 2023 and was designed as a successor to *Chandra* . Over the last year researchers have begun to utilise the greater resolution to more deeply probe the dynamics of the ICM in some clusters (Sarkar et al., 2025; XRISM Collaboration et al., 2025). Although not the focus of this thesis, X-ray observations provide key ancillary data in investigating the processes involved.

Cite telescopes

Science goals are on p. 10 of the Japanese slides for Hitomi

1.1.3 Radio

Radio observations cover several orders of magnitude, with frequencies from 10 MHz up to over 100 GHz. This massive range comes with several difficulties for astronomers, with the primary one being the angular resolution of the telescope. Angular resolution θ defines the smallest area that can be resolved given an aperture and the observing wavelength.[6] Equation 1.2 gives the formula for angular resolution in arcseconds, or $\frac{1}{3600}$th of a degree, for an aperture of diameter D and observing wavelength λ.[7]

$$\theta = 2.5 \times 10^5 \frac{\lambda}{D} \tag{1.2}$$

Radio waves range from 3 cm to 3 km, orders of magnitude larger than optical ($\sim 1\mu$m) and X-ray ($0.01-10$ nm). It is then clear that for a fixed telescope size, radio waves have much larger angular resolutions and therefore low sensitivities when compared to shorter wavelength

[6] The angular resolution of a human eye (with 20/20 vision) is 1 arcminute. *See note 7*

[7] Arcseconds (''), arcminutes ($1' = 60''$), and degrees are common units in astrophysics as declination, the celestial equivalent to latitude, is measured in these units.

radiation. In order to observe over this range several types of telescopes have been developed, including single dish telescopes, interferometer arrays, and very long baseline interferometry (VLBI).

The size of a single dish antenna is limited by both practical construction and from scientific use cases. The largest dish is the *Five-hundred-meter Aperture Spherical Telescope* (FAST) located in Guizhou China, though of the 500 metre diameter only a 300 metre area is usable at a time. Several smaller 100 metre diameter dishes exist such as the *Green Bank Telescope* in West Virginia United States, and the *Effelsburg Telescope* near Bonn, Germany. These dishes have a limited view of the sky and variable amounts of steerability due to their size, alongside hard limits on their resolution at higher radio frequencies. These restrictions developed into interferometry, the use of an array of single dish telescopes in tandem to simulate a larger dish. The *Karl G. Jansky Very Large Array* (VLA) in New Mexico United States, is comprised of 28 25-metre radio telescopes mounted on railway tracks in a Y-shaped array, observing between 73 MHz and 50 GHz. The *MeerKAT* telescope in the Meerkat National Park of South Australia contains 64 13.5-metre telescopes and operates between 1000 MHz and 10 GHz. These arrays can simulate observations by dishes several kilometres in diameter. VLBI takes this concept even further by using radio telescopes in different continents, synchronised by atomic clocks, to simulate dishes comparable to the size of the Earth. This thesis uses observations taken by the VLA, and the physics of these telescopes will be discussed in more detail in Chapter 2.

References for telescopes

Radio emission detected by these telescopes falls into the category of 'synchrotron' emission. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012; Schneider, 2014). This is due to the Lorentz force ($\vec{v} \times \vec{B}$) perpendicular to their direction of motion, which induces a helical path through the field, and therefore constant acceleration. This is analagous to brehmsstrahlung radiation, but occurs due to a magnetic interaction instead electric, which is why it is somewhat rarely called 'magnetobrehmsstralung'. A detailed discussion of the spectral distribution and features of this radiation is given in Chapter 2.

1.1.4 Gamma

The last region of interest for minihalo research is in detecting gamma radiation, or gamma-rays. This is the high energy limit for astronomy, where telescopes are designed to detect photons with energies in the MeV and GeV range.[8] These prove difficult to not only detect, due to the majority of rays being absorbed in the atmosphere, but also to accurately describe their origin (Schneider, 2014). The *Fermi Gamma-ray Space Telescope* was built specifically for this research and was

[8] Like X-ray, gamma-rays are measured by their energy rather than frequency.

launched in 2008 carrying the Large Area Telescope (LAT), with an energy range between 30 MeV and 300 GeV.

There are various sources for gamma-rays, such as Gamma Ray Bursts (GRB), which are extremely short and intense flashes. Many of these have been spatially located to coincide with bright AGNS called blazars, and rapidly spinning magnetised neutron stars called pulsars. Another source is from gamma-ray background radiation, which is expected be via neutral pion decays resulting from proton-proton collisions. Attempts to detect gamma-ray emissions connected to a clusters ICM have been ongoing as they allow us to put limits on effectiveness of proposed mechanisms to create radio halos (e.g. Reimer et al., 2003; Ackermann et al., 2010; Zandanel and Ando, 2014). Though progress has been made for giant radio halos, in particular firm constraints for the Coma cluster (Brunetti et al., 2012), minihalos have a smaller spatial extent and are much fainter, with no conclusive studies. The role of gamma rays in determining the processes creating minihalos is detailed in Chapter 2.

1.2 Diffuse Radio Emission

sources for drse

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales of giant radio halos, matching the full ICM extent, and minihalos that are contained within a clusters cool core. Relics instead reflect particles that have been re-accelerated to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Revived AGN sources are radio plasma that has been emitted from and AGN and re-energised similarly to the radio relics.

1.2.1 Production

A key question in relation to the radio halos is how they are produced. The relativistic particles producing this emission have a characteristic lifetime due to the energy losses as they travel through space, and therefore a limited diffusion length-scale in which we should be able to observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even accounting for bulk motion of the ICM and more favourable diffusion environments are well below the Mpc scales observed in many clusters (Bagchi et al., 2002). One could argue that the powerful magnetic fields supermassive black holes like M87* have and its 1.5 Mpc jet of relativistic particles could meet the length-scale criteria, but these strong fields also corral the particles like a funnel, preventing the observed radial diffusion. Two primary theories have been brought forward to explain this phenomenon, one being turbulence in the ICM causing

Citation for M87*

"re-acceleration" of once relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014). The other is inelastic collisions between relativistic cosmic ray protons and thermal protons, termed "hadronic" or secondary production (e.g. Dennison, 1980; Dolag and Enßlin, 2000; Keshet and Loeb, 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Both of these mechanisms produce similar observable features, which makes disentangling the true source much more challenging. Investigating into spatial extents and spectral features of giant radio halos has given the re-acceleration theory more credence than hadronic production (Brunetti et al., 2012), however, the debate is still ongoing for minihalos. A detailed explanation of these processes is provided in Section 2.2.3 and 2.2.4. Due to their faint nature and smaller size, observations require high sensitivity and resolution in order to accurately capture an image.

1.2.2 Minihalo detections

Talk about how many are detected, previous studies and their frequencies

1.3 This Thesis

1.3.1 Aims

The primary goal of this thesis is to calibrate, image, and analyse new radio observations of 12 minihalos at 10 GHz. These observations were performed by the *Very Large Telescope* in C and D configuration over two periods in early 2024 and early 2025. This data constitutes the first broad survey of minihalos at high radio frequencies.

This data will complement previous surveys at lower frequencies taken by the *Low-Frequency Array* (LOFAR) at 140 MHz and MeerKAT at 1.3 GHz . Processing this data will allow measurements of the minihalo flux to investigate if steepening of the spectrum occurs at higher frequency, which will assist in differentiating between the reacceleration and hadronic scenarios.

See if we have an actual reference by now

1.3.2 Work

In pursuing these aims, I downloaded the raw interferometric data from the National Radio Astronomy Observatory Archive[9] and documented the calibration and imaging process. I then performed various analyses on the final images for each minihalo to evaluate the limits and usefulness of 10 GHz observations for characterising the production mechanisms of minihalos.

This work was done in a source controlled repository[10] so that the results are reproducible and documented. Due to the size of the datasets, (uncalibrated $\sim$ 1.4 TB compressed, calibrated

[9] https://data.nrao.edu/portal/
[10] https://github.com/MoonlitMoo/Masters-2025

Add final size $\sim$ 2.5 TB, imaging $\sim$? TB), this is stored on the Storage fOr Learning And Research (SOLAR) large scale storage provided by Victoria University of Wellington.

The repository is split into two primary sections, *Data Processing* and *Image Processing*. The first section follows the calibration of the raw data and separation from observation runs into cluster specific datasets, split by observation id. The following image processing includes the cleaning and creation of the images from the trimmed datasets and the associated analysis and is split by cluster.

1.3.3 Layout

The introduction gives a brief overview for important concepts relating to minihalo physics and research.

Chapter 2 details the concept of interferometry used in the collection of the data, alongside an indepth discussion into previous studies on reacceleration and hadronic production.

Chapter 3 introduces the collected data and explains the calibration and imaging process used.

Chapter 4 explains the tools and techniques used in the analysis of the final images.

Chapter 5 details the results found for each cluster, with the context of previous studies.

Chapter 6

Chapter 7

- We want to be able to measure minihalo flux for this subset of minihalos at high radio frequencies so that we can investigate if steepening is occuring.
- This is the layout of the paper and what each chapter will cover in pursuit of said goal.

Chapter 2

Background

This chapter would be about radio interferometry, how it works to some extent and the data that is retrieved from it. Then we follow up with more specific data about radio emission.

2.1 What is an interferometer

- Limits of dish size in resolution, causes us to explore different ideas.
- How does an interferometer work.
- Bunch of dishes being interfered, delay to tune positioning, uv plane to determine coverage.
- From all of this we retrieve amp + phase measurements.

2.2 Radio emission in more detail

2.2.1 Large Radio Halos

- Use similar physics but much larger scale.
- Some basics things about previous research in differentiating production.
- Gamma ray non-detection.
- General consensus is that its re-acceleration.

2.2.2 Minihalos

- These are much smaller, so therefore much harder to observe. Difference to large radio halos.

- Similarities between host clusters (CC), general properties of spectral index etc.

2.2.3 Turbulent re-acceleration

- What is the process here.
- How manifest spectrally.
- How manifest spatially.

2.2.4 Hadronic Production

- What is the process for hadronic production
- How does this manifest spectrally and evolve. What should we expect observationally here based on simulation work.
- How does this manifest spatially and evolve, what should we see observationally.

2.2.5 Previous minihalo studies

- Generally look at lower frequencies, or a singular study for a particular minihalo.
- Want to look at more minihalos in a more holistic manner.
- Some findings from previous papers e.g. RXJ1720, MS1455+2232 in basic detail. (more detail on the per-halo discussion).

Chapter 3

Calibration and Imaging

This would be how to calibrate and the calibration results for the datasets, plus any special considerations we took.

3.1 Our observations

- We are using VLA data in X-band, what does this mean and what are the specs for the VLA.
- These observations are of list minihalos and this setup, observation proposal.
- Original source of these minihalos is the G14 paper (among others).
- Table of minihalos + observing setup.

3.2 How to calibrate a dataset

- We have the VLA calibration pipeline to remove stress from our lives.
- We have a non-neglible amount of RFI to add stress to our lives.
- A variety of steps but first pain, what is RFI and what does it cause.

3.3 Flagging data

- Initial testing showed that the pipeline flagging steps didn't work well for our data.
- AOFLAGGER worked much better and has a mostly out-of-box VLA script.
- How does AOFLAGGER work

- Integrating into a pipeline step to still use the pipeline

3.4 Calibration pipeline

- Work through each calibration step explaining what it does and why it is important.
- Final preparation for imaging is averaging down to further reduce noise.
- Notes about general trends from the calibration, with RFI in 10-12 GHz D config being bad, much of it being flagged.

3.5 Special mentions

- Either here or during following imaging section talk about the 2-3 datasets we processed slightly differently due to bad fits.
- Describe the fitting process for the custom bandpass.

3.6 Imaging

- When we make a raw image from the data, we still have a lot of artefacts from the point spread function, which comes from the beam.
- This means we need a more advanced imaging process to remove this to get a clean image. Implemented in a few different ways, we choose tclean in CASA.
- Tclean works by modelling the PSF, creating an empty model, find brightest point in residuals, model that as real, subtract model from residuals, repeat 20K times.
- This works mostly well, but we can improve results using self calibration.

3.6.1 Self calibration

- The phase calibration is the most iffy part for the main calibration, so we can make a test image and check the model seems reasonable.
- Then use that model to make small phase corrections to our dataset. Then repeat that a few times and these corrections should converge since it is circular.
- This is the final calibration steps that we do, phase corrections 20 degrees $\rightarrow$ 1 degree after a few rounds.

3.6.2 Final cleaning

- Between self calibrations this we also check the datasets and do some final manual cleaning.
- Clipping the datasets to a sensible amp range, removing spikes, SPWs where there is small amounts of data left.
- Overview of the `assess_spw` function used for this.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

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